What happens when we see the Earth as a giant system, where all the parts (including us) are linked in some way to all the other parts?

One of the ways in which scientists and philosophers have tried to understand the Earth as a complex whole is known as *the Gaia hypothesis*[1]. This was an idea put forward by the scientist James Lovelock. Lovelock said the biosphere as a whole behaves as if it is one big interconnected system.

EXTENSION TOPIC: CLOSED AND OPEN SYSTEMS

People who study systems make a distinction between a closed system and an open system. The Earth is a system that receives energy from the sun and has a certain amount of heat retained in its core. These two energy sources prevent the planet from being frozen solid. A system that receives energy from the outside is called a thermodynamically open system. All living things (like you and me, like animals and plants) are thermodynamically open because they all need energy (in the form of sunlight or food and minerals) to continue to live. But animals and plants (and the Earth) are also partially closed systems in that they have a boundary (like our skin or the bark of a tree) and they all have internal systems that organise them.

In humans, the system involves our organs, such as our lungs and heart, but it also involves the way chemicals circulate and react in our bodies, the way heat is regulated and so on. Systems that have an internal organisation (like all living things) are called organisationally closed systems.

If Earth is a dynamically open system and all plants and animals are dynamically open systems, does that make the Earth an organism? Why?

If the Earth is a dynamically open and organisationally closed system, does that make it an organism? (It has a number of things in common with all living things.)

All is process

Let's imagine for a moment that the world is not made up of objects, but is really a big and complex set of *processes* and that it is our minds that give names to things so we can make sense of the world. One way to think about this is to see the world in terms of concentrations of energy. Objects that appear to us as if they are not changing, like mountains and rocks, are very dense concentrations of energy while things that change rapidly, like clouds in a summer sky or light from the sun, are loose concentrations.

Imagine the network of electricity that flows through a city like Perth or Sydney or New York. Now imagine the network without the houses, businesses, factories, schools and hospitals. Where these things would be we see a denser concentration of energy with lines of energy entering and leaving.

1 The name 'Gaia' came about because Gaia is the ancient Greek goddess of Earth. Her name could also be spelled 'Ge', which is where we get words such as geology and geography.

Two of the key ideas in a process view of the world are Time and Change. The way we commonly understand Time is as the period between two events. Change is possibly more difficult to understand, but for there to be change, there must be time and for there to be time there must be change.

Heraclitus (6th century BCE) wrote that all the things that we perceive through the senses are in constant flux and change. He is famous for the saying that we cannot step into the same river twice. This appears in Plato's dialogue *Cratylus* (402A) as:

> Heraclitus ... says that all things are in process and nothing stays still, and likening existing things to the stream of a river he says that you can not step twice into the same river.

This is because the flowing river that we saw a minute ago has 'changed' – the stick floating downstream has moved, the water has moved, the fish have moved and so on.

- In what ways is it not possible to step into the same river twice?
- In what ways is it possible to step into the same river twice?
- Do you think that Heraclitus might have been using the river as a metaphor for change in all things?

If everything is always in constant flux (change), then the ultimate reality is change itself. Discuss. Keep in mind that change can be change in location, change in shape or form and change in what we want.

Parmenides' student Zeno of Elea[2] produced a series of paradoxes that challenged the idea of change.

Zeno's paradoxes of motion are still being used by philosophers to get students to think about the idea of change and whether change is a necessary part of reality. One way of thinking of a paradox is that it is a statement that is apparently true but which goes against our intuition about what is right. Zeno described four paradoxes:

- *The Stadium*
- *Achilles and the Tortoise*
- *The Flying Arrow*
- *The Moving Rows*

The first of Zeno's paradoxes, *The Stadium*, says that it is impossible for a runner to move from one end of a stadium to the other. The reasoning goes like this: before the runner can reach the far end, he must reach the half-way point. Before he can reach the half-way point, he must reach half-way to that point and so on. This continues to infinity. That is, the distance can always be halved, right down (in modern measures) to nanometres and smaller. There is an infinite number of points that need to be reached. And because an infinite number cannot be reached in a finite time, the stadium cannot be crossed. This paradox assumes that space is infinitely divisible.

The second paradox, *Achilles and the Tortoise*, has Achilles setting out to beat a tortoise in a race. The tortoise gets a head start. But Achilles cannot overtake the tortoise because by the time he reaches the point from which the tortoise started, the tortoise will have moved on. Whenever Achilles reaches the point where the tortoise was, the tortoise has moved on.

2. (There was another famous philosopher called Zeno the Stoic.)

The third paradox, *The Flying Arrow*, assumes that both time and space are infinitely divisible. That is, time and space can be divided into an infinity of smaller and smaller parcels. The paradox goes like this: an object is at rest when it occupies a space equal to its own dimensions. An arrow in flight occupies, at any given moment, a space equal to its own dimensions. Therefore an arrow in flight is at rest.

Zeno's paradoxes are thought experiments that attempt to show that we cannot be certain about the nature of time and space and cannot be certain that there is such a thing as change. How successful are they?

Extension investigation

Investigate Zeno's paradox of the moving rows.

All is dual

For the **Taoists** (beginning with Lao Tzu in China in the 6th century BCE) one of the most important insights 'was the realisation that transformation and change are essential features of nature', as seen in this passage from the *Chuang-tzu:*

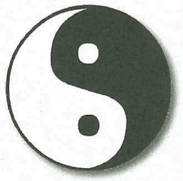

Figure 3: the Yin-Yang symbol

In the transformation and growth of all things, every bud and feature has its proper form. In this we have their gradual maturing and decay, the constant flow of transformation and change.

'The Taoists (usually pronounced Dow-ists) saw all change in nature coming from the interplay between the polar opposites yin and yang…'[3] This is commonly represented by a circular symbol that is half black and half white, where the black half has a small circle of white and the white half has a small circle of black.

What might be some of the reasons why the yin-yang symbol has white in the black half and black in the white half?

According to the Taoists, everything has a bit of yin and a bit of yang in it and yin and yang are each trying to become dominant. The dynamic relationship between yin and yang is a fundamental element of a world in constant change. Yin is an inward force and yang an outward force. A warrior in a fight is showing a lot of yang. An artist contemplating the beauty of a vase of flowers is showing a lot of yin. Yin is often associated with feminine characteristics and yang with masculine characteristics. But everything (including all males and all females) has both yin and yang forces within.

3 Capra, F. (1975). *The Tao of physics.* London: Flamingo, p. 126.

List some things that you think might be:

1. More yang than yin
2. More yin than yang

What might the ideas of yin and yang tell us about the nature of reality? For example, does it tell us that every thing has opposing forces within it?

Language and reality

What we understand about ourselves and our world we understand primarily in some form of language. Try to imagine how we might understand the world if we did not have any signs, any symbols or any words to use. Would we be able to think about the world – to understand it – if we did not have some sort of language to think with? We would not be able ask the question: what do we understand by ultimate reality? In fact, even a question as simple as 'would you like a cup of tea?' would be impossible.

Would we be humans if we did not have language? (Think back to Chapter 5 and what it means to be a person, and consider that animals without languages, such as cats and rats, can function in the world pretty adequately.)

How do we learn the concept of a 'hat'? Does the word 'hat' tell us anything about what a hat is? What would the word 'hat' tell us about an object with that name if we had not seen pictures of hats or worn a hat?

A famous example of a mistake that can happen with names comes from Abraham Lincoln, the American president. He asked: if you call a tail a leg, how many legs does a dog have? He answered his own question this way: a dog has four legs; calling a tail a leg does not make it a leg.

Mind

If we ask what ultimate reality is, we might also need to ask what it is that is doing the thinking. To be able to ask the question at all we need to be thinking. And to be thinking we need to have something to think with. The most common name for that thinking thing is a *mind.* But there are some big questions to do with the mind. For example:

- What is a mind?
- Do we need a body to have a mind? (We asked a question like this in Chapter 5.)
- Is a mind the same as a brain?
- Does the mind 'live' in the brain?
- Do non-human animals have a mind?
- Is there a real world, or is it all in our mind?

If you think, act, perceive or feel, then you have a mind. The study of the functions of minds (thinking, acting, feeling and perceiving) is called psychology. But there may be more to minds than these functions. For example, there are almost certainly subconscious processes going on.

Philosophers have been thinking about the mind since philosophy began. The early philosopher Anaxagoras (500 BC-428 BC) thought that mind (the Greek word he used was *nous*) 'arranges everything' and is the force that separates one thing from another. The mind, in effect, makes sense of the world.

How does the mind make sense of ultimate reality?

How do you start to make sense of a thought that is so big you cannot get your mind around it?

One philosopher who tried to explain this was Immanuel Kant (1724-1804).

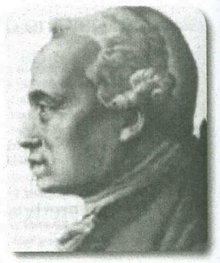

Figure 4: Immanuel Kant, German philosopher

Kant is one of the greatest philosophers of all time. His work is difficult to read, but is very important nevertheless. Kant is especially famous for what are called his three 'Critiques' *(Critique of pure reason, Critique of practical reason* and *Critique of judgement).* In the third critique, the *Critique of Judgement*, Kant distinguished between the *sublime* and the *beautiful*. Here we will touch only on the sublime. (The word 'sublime' means 'high, lofty, great, and vast' and it can refer to things in nature or in art.) For Kant, the feeling of the sublime comes to us in some cases when our reason triumphs over our imagination. And there are two types of sublime feeling: the *mathematically sublime* and the *dynamically sublime*.

We encounter the **mathematically sublime** when what we come across is experienced as large beyond measure, as unimaginably big. Our mind tries to understand this as one single thing, but it can't. Try to think about the mathematical concept of infinity: a series of numbers that can never end. Now try to think about an infinity of infinities. We cannot grasp the idea as one whole thing. We feel completely helpless until the power of our reason steps in and sort of puts a set of brackets around the idea so we can deal with it.

Try it now. We are used to thinking about infinity. Although it has no limit, our reason attempts to put limits around it. We even give it a symbol in mathematics. But how can we think about something like infinity plus one or an infinity of infinities? What is going on in your mind when you start to think about this sort of idea?

We confront the **dynamically sublime** when nature is experienced as something so powerful that we feel threatened and about to be destroyed until another idea of reason takes over and rescues us from our fear. Kant expresses the dynamically sublime in the following way:

> … thunderclouds piled up to the vault of heaven, borne along with flashes and peals; volcanoes in all their violence of destruction; hurricanes leaving desolation in their track; the boundless ocean rising with rebellious force; the high waterfall of some mighty river, and the like, make our power of resistance … a trifle … in comparison with their might. But, provided our own position is secure, their aspect is all the more attractive for its fearfulness; and we readily call [the feelings evoked by] these objects sublime, because they raise the forces of the soul above the … commonplace, and we discover within us a power of resistance … which gives us courage to be able to measure ourselves against the seeming omnipotence of nature.
>
> (*Critique of judgement*, Book II, Part B, Section 2, p. 261. Translated James Creed Meredith)

What might be some of the reasons why our own position needs to be secure for us to feel the dynamically sublime?

What could be some of the differences between the feeling of fear and the feeling of the sublime?

In what ways might the feeling of the sublime help us to understand ultimate reality?